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CHECKLIST

SMALL CRAFT - PERMANENTLY INSTALLED FUEL SYSTEMS

Ref.: EN ISO 10088:2023 (ISO 10088:2022)

Checklist_Evaluation_Module B_G en250115

Watercraft manufacturer:	
Watercraft model name:	

	Subject to check	Clause	Requirements	Checked ?
1	Fuel type.		[Petrol / Diesel / both]	
2	Petrol engine compartments and petrol tank compartments shall have ventilation and ignition protection in accordance with ISO 11105 and ISO 8846.	4.1.4	[Yes / NA ?]	
3	If petrol, the only outlets for drawing fuel from the fuel system are plugs in petrol filter bowls for the purpose of servicing filter.	4.1.5	[Yes / NA ?]	
4	If diesel, the only outlets for drawing fuel from the fuel system are plugs or valves in diesel filter bowls for the purpose of servicing filter.	4.1.5	[Yes / NA ?]	
5	If petrol, each metal or metallic plated component of fill system and tank is grounded. Resistance measured, see testing checklist.	4.1.6	[Yes / NA ?]	
6	Grounding wires are not clamped between a hose and its clamps.	4.1.6	[Yes / NA ?]	
7	If copper-base alloy fittings are used for aluminium tanks: Protection by a galvanic barrier ?	4.1.10	[Yes / NA ?]	
8	Means to determine fuel quantity is provided.	4.1.11	[Yes ?]	
9	Fuel system is permanently installed.	4.3.1	[Yes ?]	
10	All parts, except small connectors, fittings and short sections of flexible hoses, are independently supported.	4.3.1	[Yes ?]	
11	All components intended to be operated or observed during normal operation of the craft, or for emergency purposes, are readily accessible.	4.3.2	[Yes ?]	
12	All fittings and connections are at least readily accessible, or accessible through an access panel, port or hatch. Tanks need not be accessible for removal.	4.3.2	[Yes ?]	
13	Clearance between petrol fuel tank and combustion engine > 100 mm.	4.3.3	[Yes / NA ?]	
14	Clearance between petrol tank and exhaust components with a temperature exceeding 90°C is > 250 mm if no thermal barrier is provided.	4.3.4	[Yes / NA ?]	
15	Fuel tank(s) and components of petrol fuel systems are not installed directly above batteries unless the batteries are protected against fuel leakage.	4.3.6	[Yes / NA ?]	
16	Minimum inside diameter of the fill pipe system is 28,5 mm.	5.1.1	[Yes ?]	
17	Minimum inside diameter of fuel filling hoses is 38 mm.	5.1.1	[Yes ?]	
18	Fuel filling hoses in the engine compartment are of fire resistant type A1, A2 or A15 as per ISO 7840.	5.1.2	[Yes / NA ?]	

compliant: Yes or √ not applicable: NA follow up on variation report: Rpt or X

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19	Fuel fill hoses outside the engine compartment are of type A1 or A2 as per ISO 7840, or of type B1, B2 or B15 as per ISO 8469.	5.1.2	[Yes / NA ?]	
20	Fuel filling lines are self-draining to the tank, craft being in static floating position.	5.1.3	[Yes ?]	
21	Distance between compartment ventilation openings and fuel fill openings are at least 400 mm. Acceptance if craft's coaming, superstructure or hull creates a barrier to prevent fuel vapour entering the craft through ventilation opening.	5.1.5	[Yes ?]	
22	**Label:** Fuel filling point marked with “petrol” or “diesel” and/or with a symbol as per ISO 11192. **Affixed:** on the fuel filling point.	5.1.6	[Yes ?]	
23	Each fuel tank has separate vent line.	5.2.1	[Yes ?]	
24	Vent hoses in the engine compartment are of fire resistant type A1, A2 or A15 in accordance with ISO 7840.	5.2.2	[Yes / NA ?]	
25	Vent hoses outside engine compartment are of type A1, A2 or A15 as per ISO 7840, or type B1, B2 or B15 as per ISO 8469.	5.2.2	[Yes / NA ?]	
26	Vent lines are self-draining when craft in static floating position.	5.2.5	[Yes ?]	
27	Distance between compartment ventilation openings and fuel vent openings is at least 400 mm. Acceptance if craft's coaming, superstructure or hull creates a barrier to prevent fuel vapour entering the craft through ventilation opening.	5.2.6	[Yes ?]	
28	Vent line arrangement minimizes the intake of water without restricting the release of vapour or intake of air and does not allow the vapour overflow to enter the craft.	5.2.7	[Yes ?]	
29	Vent-line termination or gooseneck in the vent-line routing is located above the heeled waterline of the craft.	5.2.8	[Yes ?]	
30	Vent lines on all petrol fuel installations incorporate a flame arrester unless the ventline is included as part of a combined fill and vent fitting.	5.2.9	[Yes / NA ?]	
31	Components in vent lines that can capture fuel must be fire-tested as per 4.1.9.	5.2.10	[Yes / NA ?]	
32	Metal fuel distribution and return lines are of seamless annealed copper or copper–nickel or equivalent metal with nominal wall thickness of at least 0,8 mm. Aluminium lines may be used for diesel fuel.	5.3.1	[Yes / NA ?]	
33	Rigid fuel distribution and return lines are connected to the engine by a flexible hose section.	5.3.2	[Yes / NA ?]	
34	Support of rigid fuel distribution and return lines are provided within 100 mm of the connection to the metal supply line on the rigid side of the connection.	5.3.2	[Yes / NA ?]	
35	Connections in rigid fuel distribution or return lines are made with efficient screwed, compression, cone, brazed or flanged joints.	5.3.3	[Yes / NA ?]	
36	Flexible fuel hoses are used where relative movement of the craft structures supporting the fuel lines would be anticipated during normal operating conditions.	5.3.4	[Yes / NA ?]	
37	Flexible fuel hoses are accessible for inspection and maintenance.	5.3.5	[Yes ?]	

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38	Petrol distribution are of fire-resistant type A1, A2 or A15 as per ISO 7840, except hoses entirely within splash well at stern of craft connected directly to outboard engine by type B1, B2 or B15 hoses as per ISO 8469 or A1, A2 or A15 hoses as per ISO 7840.	5.3.6	[Yes / NA ?]	
39	Diesel-fuel distribution and return hoses are of fire-resistant type A1 or A2 as per ISO 7840.	5.3.7	[Yes / NA ?]	
40	Fuel lines are properly supported and secured to craft structure above bilge water level, unless specifically designed for immersion or protected from the effects of immersion.	5.3.8	[Yes ?]	
41	There are no joints in fuel distribution and return pipes or hoses other than those required to connect required fuel-line components, e.g. filters and bulkhead connections.	5.3.9	[Yes ?]	
42	Fuel distribution lines to petrol engine(s) prevent fuel siphoning out of the tank following a failure in the system by: - routing all parts of the fuel line above the level of the tank top; or - fitting an anti-siphoning valve or manual or electrical operated valve as close as practical to the fuel tank.	5.3.10	[Yes / NA ?]	
43	Fuel distribution lines to diesel engine(s) prevent fuel siphoning out of the tank following a failure in the system by: - meeting 5.3.10, or - fitted with a manual shut-off valve.	5.3.11	[Yes / NA ?]	
44	Diverting valves in diesel return lines ensure that the return line flow is not restricted.	5.3.12	[Yes / NA ?]	
45	Fuel hoses are secured to the pipe, spud or fitting by metal hose clamps or are equipped with permanently attached end fittings.	5.4.1	[Yes ?]	
46	Pipes, spuds (except fuel-tank spud) or other fittings for hose connection with hose clamps have a bead, flare, series of annular grooves or serrations.	5.4.2	[Yes ?]	
47	Spuds or other fittings for hose connection with hose clamps have a nominal outer diameter being the same as the nominal inner diameter of the hose.	5.4.3	[Yes ?]	
48	Hose connections designed for a clamp connection have a spud at least 25mm long.	5.4.4	[Yes ?]	
49	Hose connections having a nominal diameter of more than 25mm shall have two hose clamps. The spud is at least 35mm long.	5.4.5	[Yes / NA ?]	
50	Spuds for hose connection are free from sharp edges.	5.4.6	[Yes ?]	
51	Hose clamps are of CrNi 18-8 stainless steel, or equivalent, and reusable.	5.4.7	[Yes ?]	
52	Clamps depending solely on spring tension are not used.	5.4.7	[Yes ?]	
53	Nominal clamp band width is at least 8 mm for nominal outside hose diameters up to and including 25 mm and at least 10mm for bigger hoses.	5.4.7	[Yes / NA ?]	
54	Clamps are installed to fit directly on the hose and do not overlap each other.	5.4.8	[Yes ?]	
55	Clamps are installed behind the bead, if any, or fully on the serrations on spuds with at least one clamp width from the end of the hose.	5.4.8	[Yes ?]	

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56	Manually operated valves have positive stops in the open and closed positions or clearly indicate their open and closed positions.	5.5.1	[Yes / NA ?]	
57	The integrity and tightness of a valve does not depend solely on spring tension.	5.5.2	[Yes ?]	
58	Threaded valve housing covers that can be exposed to an opening torque when the valve is operated are secured against unintentional opening by a device that can be reused.	5.5.3	[Yes / NA ?]	
59	If transparent sight gauge is installed on diesel tank, it is mounted as close as practical to the tank, minimizing the risk of physical damage. It has a self-closing device on the bottom and a valve (not self-closing) at the top.	5.5.4	[Yes / NA ?]	
60	Petrol fuel systems is equipped with a fuel filter. This maybe fitted on the engine.	5.6.1	[Yes / NA ?]	
61	Diesel fuel systems is equipped with at least one fuel filter and one water separator or being combined into one device.	5.6.2	[Yes / NA ?]	
62	Each filter is independently supported on the engine or craft structure.	5.6.3	[Yes ?]	
63	All system components that fulfil ISO 10088 shall be marked or labelled: - manufacturer's name or trademark; - ISO 10088 - fire resistant: - type of fuel or fuels for which the component is suitable.	5.7	[Yes ?]	

The following questions shall be filled in by the watercraft manufacturer and appropriate documentation shall be submitted to the inspector for verification.

	Subject to check	Clause	Requirements	Checked ?
65	Individual components of the fuel system, and the fuel system as a whole, shall be designed to withstand the combined conditions of pressure, vibration, shocks, corrosion and movement encountered under normal operating conditions and storage.	4.1.1	[Yes ?]	
66	Each component and system as whole operates throughout ambient temperature range of -40 °C and +80 °C.	4.1.2	[Yes ?]	
67	Each component and system as whole is resistant to deterioration to all liquids or compounds with which it may come into contact.	4.1.3	[Yes ?]	
68	If petrol, each metal or metallic plated component of fill system and tank is grounded with less resistance than 1 ohm.	4.1.6	[Yes / NA ?]	
69	Fuel filling systems is designed to avoid spillage of fuel during refuelling to the rated capacity and tested in accordance with 4.2.3	4.1.7	[Yes ?]	
70	Provision shall be made to prevent fuel overflow from the vent opening while refuelling from entering the interior of the craft or the water. Note: A substance is “entering the interior of the craft”, when it gets into a place being inside the surface of the watercraft. This can be the cabin or a similar place not being open to the atmosphere having one or more closing appliances used to cover an opening in the cockpit, hull or superstructures.	4.1.8 4.2.2	[Yes ?]	

compliant: Yes or √ not applicable: NA follow up on variation report: Rpt or X

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71	All fuel system components in the engine compartment, except permanently installed fuel tanks withstanding a 2,5 min fire test as specified in Annex CISO 7840 (individually or as installed). Fasteners supporting metal fuel lines are exempted.	4.1.9	[Yes / NA ?]	
72	The whole fuel system passes after installation the pressure test as specified in Annex A.	4.2.1	[Yes ?]	
73	Blow back test conducted.	4.1.7 4.2.3	[Yes ?]	
74	The fuel filling system shall be designed so that accidental fuel spillage does not enter the interior of the craft during fuelling, when it is in its static floating position. Note: “entering the interior of the craft” means when a substance gets into a place being inside the surface of the watercraft. This can be the cabin or a similar place not being open to the atmosphere having one or more closing appliances used to cover an opening in the cockpit, hull or superstructures.	5.1.4	[Yes ?]	
75	The inside diameter of the ventilation pipe is 11 mm (= 95 sqmm) or ventilation opening designed to prevent tank pressure from exceeding 80% of max. test pressure as marked on the tank label.	5.2.3	[Yes ?]	
76	Vent lines do not have valves other than those that permit free flow of air and prevent flow of liquid (fluid) both in and out of the tank.	5.2.4	[Yes ?]	
77	Vent-line components in engine compartments, able to capture fuel, fulfil test requirements of 4.1.9.	5.2.10	[Yes ?]	
78	Pressure testing: a test report according to the normative Annex A is provided.	Annex A	[Yes ?]	
79	Note: Annex B provides informative information about methods and tests for controlling emissions of petrol fuel systems. Documentation is voluntary.			

Comments:

compliant: Yes or √ not applicable: NA follow up on variation report: Rpt or X